
Productivity and Selection of Dragline

1. PRODUCTIVITY CALCULATION FOR DRAGLINE

1.1. Dragline Cycle Time

Dragline is also a cyclic equipment used for handling of last overburden bench just above the coal seam (last interburden bench above the deep most coal seam in case of multi-seam deposit). A dragline, in its normal operation, sits on the top of a bench and excavates below its crawler level. It transports and dumps its own load through direct casting.

Dragline working cycle consists of six components - (i) **digging** through the blasted bench material (or the in-situ face) loading the bucket, (ii) **hoisting** of the bucket, (iii) **forward swing** of the loaded bucket over the dumping area, (iv) **dumping** of the bucket load, (v) **return swing** of the empty bucket to the face area and (vi) **lowering** the bucket at bench floor level to start the next cycle. However, during a cycle of operation these six components overlap each other as explained below.

Digging	Forward swing	Return swing
Hoisting	Dumping	Lowering

For draglines two cycle times are defined – stop watch cycle time and actual (average) cycle time. The stopwatch cycle time is the time required by a dragline to complete a working cycle for 90° swing under ideal conditions and this is quoted in manufacturer's reference. The actual cycle time is more than the stop watch cycle time as it involves number of delays and occasional bad passes. Determination of actual cycle time is best done through time study over large number of working cycles under actual field conditions.

The actual cycle time (t_c) depends on the required angle of swing, the bench height, the condition of material being handled, the machine itself and the operator's efficiency. The swing angle is approximately 90° for simple side casting and extended bench method, but it may vary from 45° to 180° depending on the working method (upto 180° swing in pull-back method and 45° to 90° swing while working in vertical tandem). Lower the swing angle, lower will be the value of t_c . Generally t_c varies from 55 seconds to 65 seconds for simple side casting.

1.2. Dragline Productivity

Dragline productivity is generally expressed in terms of m^3 (bank)/hour.

Theoretical productivity (Q_{th}) may be given as

$$Q_{th} = B_c \times B_f \times \frac{100}{100 + S} \times \frac{3600}{t_c \times F_s} \quad m^3 \text{ (bank)/hour}$$

However, the actual or effective productivity (Q_{eff}) is less than the theoretical productivity and is given by

$$Q_{\text{eff}} = B_c \times B_f \times \frac{100}{100 + S} \times \frac{3600}{t_c \times F_s} \times E \quad \text{m}^3 (\text{bank})/\text{hour}$$

Where,

- B_c is the bucket capacity in m^3
- B_f is the bucket fill factor
- S is the % Swell of the spoil material
- t_c is the actual cycle time in sec.
- F_s is the factor of swing
- E is the overall utilization factor.

When the material in bank is directly excavated without prior blasting, *bucket fill factor* (B_f) becomes a function of the material characteristics. But in case of excavation of blasted rock, B_f primarily depends on the dragline bucket size and the average lump sizes of the broken material. As the ratio of the bucket size to the average lump size increases, B_f also increases. In practical situations B_f generally has a value between 0.75 and 0.85.

Percentage swell (S) is primarily a function of material characteristics, but this is also influenced by the degree of fragmentation in case of blasted rock. Higher the degree of fragmentation (i.e. Lower the average fragment size), S will be more, limited to a maximum value that is dependent on the material characteristics. Generally for coal measure rocks S has a value around 20.

Factor of swing (F_s) is a cycle time correction factor to take into account the angle of swing that the dragline has to make for the given method of working.

The overall utilization factor (E) is the ratio of the actual working hours to the scheduled shift hours (= actual working hours + maintenance hours + idle hours). So, E depends on the maintenance hours required and the idle hours corresponding to particular working hours. In India, the average value of E is about 0.7 for walking draglines.

2. SELECTION OF DRAGLINE FOR A GIVEN SITUATION

Dragline is highly productive but very costly equipment, and therefore, its optimum selection is of great importance from the point of view of economy of operation of the project.

Selection of dragline for a particular mine is a two-step procedure. First one is the technical selection, in which, a number of models (from different manufacturers) are selected on the basis of job requirements. This is followed by the selection of the optimum one on the basis of economic considerations.

For the technical selection, the required bucket capacity (B_c) is estimated from the handling requirement followed by the estimation of reach (R) from the side cast diagram, and then the maximum usefulness factor (MUF) is calculated from B_c and R . This MUF is used to select the models that may be considered for selection on the basis of economic considerations.

2.1. Estimation of Required Bucket Capacity (B_c)

For estimation of B_c , first the handling requirement per hour is estimated as follows.

$$\text{Handling Requirement} = (P \times SR) / (365 \times 24) \text{ m}^3 (\text{bank})/\text{hour} \dots\dots(i)$$

Where,

P is the required production of coal per annum in te.

SR is the stripping ratio in m^3/te .

The effective hourly capacity of a dragline is given by

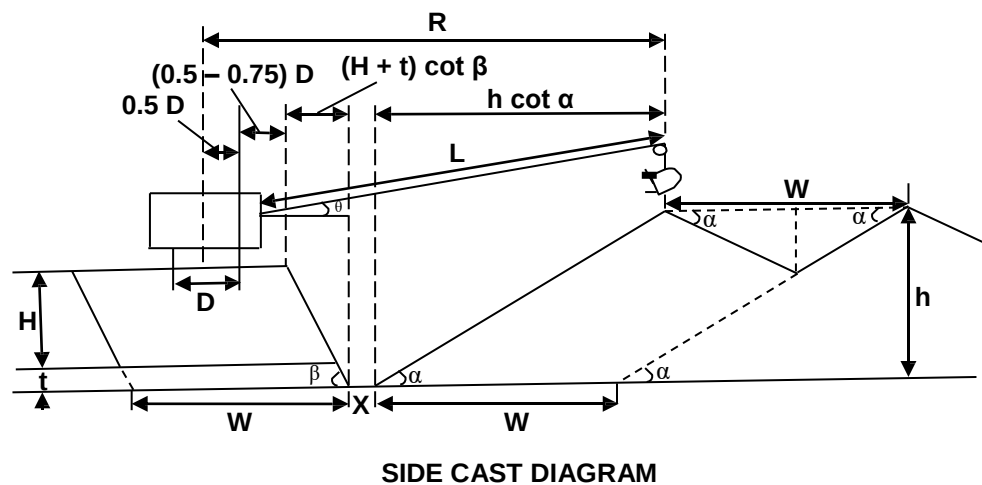
$$Q_{\text{eff}} = B_c \times B_f \times \frac{100}{100 + S} \times \frac{3600}{t_c \times F_s} \times E \text{ m}^3 (\text{bank})/\text{hour} \dots\dots(ii)$$

The required bucket capacity to fulfill the handling requirement may be estimated from (i) and (ii) as given below.

$$B_c = \frac{(P \times SR) / (365 \times 24)}{B_f \times (100 / (100 + S)) (3600 / t_c) \times E} \text{ m}^3$$

2.2. Estimation of Reach (R) and Boom Length (L)

The required reach is estimated from the side cast diagram (also known as balancing diagram or pit profile).



Where,

H is the height of overburden bench

W is the width of cut

α is the angle of slope of spoil heap

h is the height of spoil heap

θ is the boom angle

X is the clearance between the highwall toe and the spoil heap toe

t is the thickness of coal seam

β is the highwall slope angle

D is the diameter of tub (base) of dragline

L is the length of boom

S is the percent of swell of the material

$$R = (1 \text{ to } 1.25) \times D + (H + t) \times \cot \beta + h \times \cot \alpha + X$$

$$\text{As } h = H \times \left(1 + \frac{S}{100}\right) + \left(\frac{W}{4}\right) \tan \alpha + X$$

$$R = (1 \text{ to } 1.25) \times D + (H + t) \times \cot \beta + H \times \left(1 + \frac{S}{100}\right) \times \cot \alpha + \left(\frac{W}{4}\right) + X$$

$$\text{Again, } L \cos \theta \approx R - 0.75 D$$

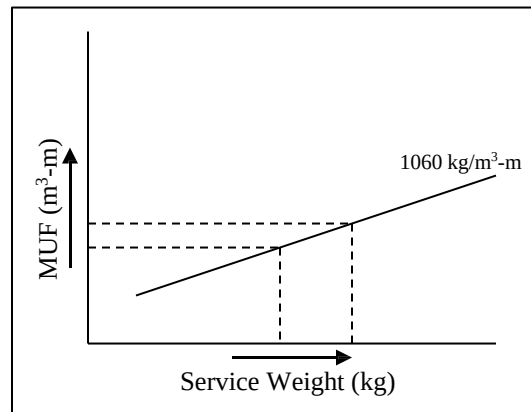
Considering the value of θ to be 35° to 39° (which is most common) $L \approx R$.

2.3. Maximum Usefulness Factor (MUF)

It is the measure of stability of the machine, and it is given by,

$$\text{MUF} = B_c \times R \approx B_c \times L$$

The MUF of a machine depends on its service weight (i.e., machine weight + weight of ballast). If the service weight and the corresponding MUF of draglines are plotted (as shown), it will be found that a line can be drawn so that all the stable machines will fall between the X-axis and the drawn line. Average slope of this line is approximately $1060 \text{ kg/m}^3\text{-m}$ for walking draglines.



After the required bucket capacity and the reach are estimated for the given application, minimum MUF required is calculated. This is followed by calculation of minimum required service weight of the machine from the service weight - MUF relationship curve. Thereafter few models from different manufacturers reference having service weight more than the minimum required service weight is selected, from which the optimum one is chosen based on the economic considerations.

3. GENERAL CONSIDERATIONS FOR SELECTION OF A DRAGLINE

3.1. Technical Considerations

Material

1. Material type
2. Material condition (in-situ or fragmented)
3. Material density (bank and loose)
4. Percent of swell of the material

Operating Conditions

5. Depth of excavation/bench height
6. Maximum dumping height
7. Minimum dumping reach (from pit profile)
8. Area of workplace

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9. Bearing quality of the working place (walking pressure-1.5 kg/cm², working pressure – 0.8 kg/cm²)
 10. Weather condition
 11. Estimated average swing angle
 12. Estimated overall utilization factor
 13. Estimated bucket fill factor
 14. Overburden/interburden handling requirement (from production requirement)

Equipment Selection

15. Estimate average cycle time (from 5 & 11)
16. Estimate the required bucket size (from 4,12,13, 14 & 15)
17. Estimate the required boom length (from 7)
18. Estimate the minimum MUF required (from 7 &16)
19. Select dragline from manufacturer reference (using 18)

Verification of Equipment Choice

20. Manufacturer name and model
21. Machine weight, service weight and MUF of the selected model.
22. Optimum depth of cut for the selected model.
23. Calculate percent of optimum depth (from 5 & 22)
24. Cycle time for 90o swing and optimum depth of cut for the selected model
25. Estimate average cycle time (from 11,23 & 24)
26. Calculate the bucket size required (from 4,12,13, 14 & 25)
27. Calculate the minimum MUF required (from 17 & 26)
28. Compare the MUF required (27) with MUF available (21) and make the final selection.

3.2. Economic Considerations

1. Machine price
 - (i) Base machine
 - (ii) Freight
 - (iii) Ballast
 - (iv) Erection
2. Depreciation (from machine life)
3. Interest
4. Insurance
5. Taxes
6. Operating and maintenance costs
 - (i) Wages of the machine crew
 - (ii) Power costs
 - (iii) Cost of consumables
 - (iv) Maintenance costs.